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Abstract: This paper argues that the century-long impasse in Indus script studies reflects two compounding failures: definitional (the term “script” has never been formalized) and methodological (statistical analysis alone cannot resolve the question). The exchange between Rao et al. (2009) and Sproat (2010) demonstrated that entropic methods can neither confirm nor exclude the linguistic hypothesis. We formalize three hypotheses—linguistic writing, administrative registration code, and hybrid system—and construct a Decision Framework comprising five testable predictions that extend the methodological system to incorporate archaeological evidence. The existing statistical evidence—including sequence uniqueness rates of 98.3%, edge-peaked positional entropy, numeral clustering, and near-duplicate analysis—is evaluated against all three hypotheses. The framework is generalizable to other undeciphered or contested sign systems.

What Would Prove It?

Definitional Indeterminacy, Methodological Undecidability, and a Decision Framework for the Indus Sign System

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Abstract

The Indus sign system, attested on approximately 4,000 inscribed objects from the Harappan civilization (c. 2600–1900 BCE), remains undeciphered after a century of scholarship. This paper argues that the impasse reflects two compounding failures. The first is definitional: the central terms of the debate—“writing,” “script,” “linguistic”—have never been formalized with sufficient precision to admit resolution. The provability of any answer varies with the definition chosen; the field has been debating a question that has not yet been posed with adequate precision. The second failure is methodological: the question of the system’s nature is undecidable within the framework of statistical analysis alone. The exchange between Rao et al. [2009] and Sproat [2010] demonstrated that entropic methods can neither confirm nor exclude the linguistic hypothesis—a symmetry that signals the framework’s insufficiency rather than the question’s unanswerability. This paper addresses both failures. We formalize three precisely defined hypotheses—linguistic writing, administrative registration code, and hybrid system—and construct a Decision Framework comprising five testable predictions that extend the methodological system to incorporate archaeological evidence. The existing statistical evidence—including sequence uniqueness rates of 98.3%, edge-peaked positional entropy, numeral clustering, and near-duplicate analysis—is evaluated against all three hypotheses. The framework is generalizable to other undeciphered or contested sign systems.

Keywords: Indus script, undecidability, definition-dependent provability, decision framework, Harappan civilization, administrative codes, undeciphered writing systems, computational linguistics

1 Introduction

For nearly a century, scholars have debated whether the signs inscribed on Harappan seals, tablets, and other objects constitute a writing system encoding spoken language. The debate has produced a substantial literature but no consensus. Proponents of the linguistic hypothesis point to the system’s combinatorial structure, its conditional entropy profile, and parallels with known scripts [Parpola, 1994, Rao et al., 2009, Yadav et al., 2010]. Proponents of the non-linguistic hypothesis point to the extreme brevity of inscriptions, the absence of longer texts, and the failure of all phonetic decipherment attempts [Farmer et al., 2004, Sproat, 2010]. As Sproat [2023] has argued, the boundary between linguistic and non-linguistic notation is less categorical than typically assumed: nonlinguistic systems can possess rich, complex structure. A third position—that

the signs encode structured administrative information without representing spoken language—has been advanced by Mukhopadhyay [2019], Mukhopadhyay [2023], Pollard [2025], and Kriger and Hunt [2026].

The standard diagnosis of the impasse is that we lack sufficient evidence. This paper proposes a different diagnosis: the impasse reflects two compounding epistemological failures that have not been distinguished from each other.

The first failure is **definitional**. The central question—“Is the Indus sign system a script?”—has never been posed with formal precision. The term “script” admits multiple admissible definitions, and the provability of any answer varies with the definition chosen. A statement that varies in provability across admissible definitions is not undecidable; it is underdefined.

The second failure is **methodological**. Statistical methods have been deployed to argue both for and against the linguistic hypothesis. The exchange between Rao et al. [2010] and Sproat [2010] exposed the limits of entropy-based approaches from both sides: entropic similarity to linguistic systems does not prove linguistic status, and entropic dissimilarity from random sequences does not exclude non-linguistic structure. The symmetry is exact, and it signals that the methodological system of statistical analysis alone lacks the resources to resolve the question.

This paper addresses both failures. Section 2 analyzes the definitional indeterminacy. Section 3 replaces the undefined binary with three precisely specified hypotheses. Section 4 evaluates existing statistical evidence. Section 5 constructs a Decision Framework—five tests whose outcomes differentially favour one hypothesis over the others. Sections 6 and 7 address specific objections. Section 8 formalizes the argument’s structure. Section 9 discusses generalizability.

2 Definitional Indeterminacy

2.1 Provability as a Function of Definition

Provability is a function of definition. For any statement whose formulation depends on terms admitting multiple admissible definitions, the provability status of that statement varies with the definition chosen. This is not a claim about the relativity of truth; it is a claim about the conditions under which proof becomes applicable. A statement that varies in provability across admissible definitions is not undecidable—it is underdefined. The persistent disagreement in Indus studies is an instance of this phenomenon.

2.2 Four Definitions of “Script”

The term “script” in the Indus debate admits at least four admissible definitions, each yielding a different answer to the question “Is the Indus system a script?”:

W₁ (Broad Semiotic): Any system of conventional signs with internal combinatorial structure used for communication. Under W₁, the Indus system trivially qualifies.

W₂ (Phonological): A system whose signs encode the phonological units of a specific spoken language. Under W₂, no successful decipherment has demonstrated this. The system does not qualify on current evidence.

W₃ (Logographic): A system whose signs represent words or morphemes without necessarily encoding phonology. Under W₃, the question remains open.

W₄ (Functional-Administrative): A system of signs used to record, regulate, or administer information, whether or not it encodes spoken language. Under W₄, the system qualifies if the administrative hypothesis is correct.

The persistent disagreement in the field is not a disagreement about evidence. It is a disagreement about definitions operating below the surface of the debate. Scholars who assert that the Indus system is a script typically operate with W₁ or W₃; those who deny it typically operate with W₂. Both are correct relative to

Table 1: The provability function for “Is the Indus system a script?” across four definitions.

Definition	Content	Answer	Status
W_1 Semiotic	Conventional signs with structure	Yes	Trivially true
W_2 Phonological	Encodes phonological units	No	Undemonstrated
W_3 Logographic	Signs represent words	Open	Indeterminate
W_4 Functional	Administrative recording	Cond.	Depends on H_2

their definitions.

2.3 Dissolving the Binary

As Sproat [2023] has argued from a comparative perspective, sign systems occupy a spectrum from fully linguistic to fully non-linguistic, and the boundaries between categories are not always sharp. The traditional binary—“writing” versus “not writing”—is an artifact of definitional imprecision. The productive question is not “Is this writing?” but “What is the structure of this system, and what function did it serve?” This question can be formalized and evaluated against evidence.

3 Three Hypotheses

To escape the definitional impasse, we replace the binary question with three precisely defined structural hypotheses. Each specifies a different functional architecture for the sign system, generating distinct testable predictions.

3.1 H_1 : Linguistic Writing

The signs encode natural language—logographically, syllabically, or through some combination. Structural predictions: (a) inscription length correlates with informational complexity of context; (b) morphological or syntactic regularities reflect the grammar of the encoded language; (c) signs combine in word-level or phrase-level patterns; (d) weak or no correlation between animal motif and sign sequences; (e) individual signs repeat within inscriptions at rates comparable to known scripts.

3.2 H_2 : Administrative Registration Code

The signs encode structured identifying information—personal or institutional identity, clan affiliation, administrative status—without representing spoken language. Structural predictions: (a) the vast majority of sign sequences are unique; (b) strong positional structure with fixed elements at edges and variable elements in the middle; (c) numeral signs cluster at specific positions; (d) numeral distribution deviates from Benford’s law; (e) near-duplicate pairs exist, differing in the variable middle zone; (f) inscription length is independent of contextual complexity; (g) animal motif correlates strongly with sign prefixes; (h) individual signs rarely repeat within an inscription.

3.3 H_3 : Hybrid System

Some signs function as logographs while the overall sequence structure serves an administrative function. Roman imperial inscriptions provide a precedent: the Roman alphabet is used formulaically (*IMP CAES*

DIVI F AVGVSTVS), with each position encoding a fixed category rather than composing free prose. Structural predictions: (a) signs at fixed positions show semantic clustering with the animal motif; signs in the variable middle do not; (b) some signs appear in both fixed and variable positions; (c) conditional entropy is intermediate between linguistic and random baselines—which is what Rao et al. [2009] demonstrated.

4 Existing Statistical Evidence and Its Limits

4.1 Sequence Uniqueness

In a corpus of 179 Mohenjo-Daro seals, 98.3% of sign sequences are unique [Kriger and Hunt, 2026]. Strongly predicted by H_2 ; consistent with H_3 ; compatible with but not naturally explained by H_1 .

4.2 Positional Entropy Profile

The entropy profile shows low entropy at positions 1 and 5, high entropy at positions 2–4. Strongly predicted by H_2 (fixed categorical markers at edges, variable code in the middle). H_1 would predict a flatter profile.

4.3 Numeral Distribution

Stroke-numeral signs constitute 15–17% of all signs and cluster at the penultimate position. The digit “2” accounts for 58–67% of all numeral tokens, drastically violating Benford’s law [Kriger and Hunt, 2026]. Strongly predicted by H_2 ; inconsistent with H_1 if the numerals represent quantities.

4.4 Near-Duplicate Pairs

Eight pairs of seals differ by exactly one sign, with the difference never at the first or last position [Kriger and Hunt, 2026]. Predicted by H_2 ; under H_1 , the restriction to the middle zone would require explanation.

4.5 Intra-Inscription Non-Repetition

Individual signs very rarely repeat within a single inscription—a robust finding first noted by Farmer et al. [2004] and consistently emphasized in subsequent work [Sproat, 2010, 2023]. Under H_2 , this follows from combinatorial selection without replacement. Under H_1 , it is anomalous: most writing systems exhibit substantial within-text sign repetition.

4.6 Seal Wear

Gokhale and Ameri [2026] present agent-based modelling evidence that many seals show minimal wear, inconsistent with repeated stamping. Compatible with carried identification objects. However, the physical function of the object is logically independent of the sign system’s nature: Mesopotamian seals included true writing and were used as seals.

4.7 Summary

The evidence is consistently compatible with H_2 and H_3 while creating multiple anomalies for H_1 . But no finding definitively excludes H_1 , and no finding uniquely supports H_2 over H_3 . The methodological system of statistical analysis alone cannot resolve the question. This is the symmetry that the Rao–Sproat exchange exposed: the same framework that fails to prove H_1 also fails to disprove it.

Table 2: Evidential status of existing findings.

Finding	H ₁ Ling.	H ₂ Code	H ₃ Hybrid
98.3% unique sequences	Compatible	Str. predicted	Compatible
Edge-peaked entropy	Weakly comp.	Str. predicted	Compatible
Numeral clustering	Anomalous	Str. predicted	Compatible
Benford violation	Anomalous	Str. predicted	Compatible
Near-duplicates in middle	Req. expl.	Predicted	Compatible
Intra-text non-repetition	Anomalous	Predicted	Constraining
Minimal seal wear	Neutral	Compatible	Neutral

5 The Decision Framework: Five Tests

If the question is undecidable within the current methodological system, and if definitional indeterminacy has been resolved by specifying three precise hypotheses, the remaining task is to extend the system. The Decision Framework specifies five tests whose outcomes differentially favour one hypothesis. Two can be performed with existing digital corpora; three require archaeological collaboration.

5.1 Test 1: Inscription–Context Correlation

Question: Does inscription length correlate with the informational complexity of the archaeological context?

Predictions: H₁: positive correlation. H₂: no correlation (fixed-format codes). H₃: weak correlation.

Status: Requires compilation of provenance data from excavation reports. Archaeological collaboration is indispensable.

5.2 Test 2: Animal Motif–Sign Prefix Correlation

Question: Do specific animal motifs (unicorn, zebu, elephant, rhinoceros) correlate with specific sign prefixes?

Predictions: H₂: strong correlation (both encode the same categorical variable). H₁: weak or no correlation. H₃: moderate, position-dependent.

Status: Testable immediately using the ICIT database [Wells and Fuls, 2010–2023]. A significant animal–prefix correlation would be strong evidence against pure H₁.

5.3 Test 3: Cross-Site Format Consistency

Question: Does the positional structure remain consistent across geographically distant sites?

Predictions: H₂: high consistency (uniform format). H₁: regional dialectal variation. H₃: formulaic elements consistent, variable elements regionally varied.

Status: Feasible with existing data. Yadav et al. [2010] noted cross-corpus bigram consistency but did not test positional entropy by site.

5.4 Test 4: Commodity–Inscription Independence

Question: When inscribed objects co-occur with specific commodities, do different inscriptions appear with the same commodity?

Predictions: H₂: independent (different owners, same commodity). H₁: correlated (inscriptions describe contents). H₃: partially correlated.

Status: Requires systematic compilation of provenance data. This is where archaeological collaboration is most critical.

5.5 Test 5: Prefix Distribution Across Sites

Question: Is there evidence for coordinated prefix allocation across sites?

Predictions: H₂: site-specific or coordinated allocation. H₁: distribution reflecting linguistic categories. H₃: dual structure.

Implications: The Harappan civilization shows remarkable standardization—weights, measures, brick ratios, urban planning—across vast distances, consistent with distributed administrative protocols.

5.6 The Decision Matrix

Table 3: Predictions of each hypothesis for the five tests.

Test	H ₁ Prediction	H ₂ Prediction	H ₃ Prediction
1. Length–context	Pos. correlation	No correlation	Weak correlation
2. Animal–prefix	Weak/none	Strong corr.	Moderate; pos.-dep.
3. Cross-site format	Regional variation	High consistency	Mixed
4. Commodity indep.	Correlated	Independent	Partially corr.
5. Prefix distribution	Ling. distributed	Admin. structured	Dual structure

No two hypotheses generate identical predictions across all five tests. A clear result on even two or three would substantially narrow the hypothesis space.

6 The Roman Inscription Objection

A natural objection: Roman imperial inscriptions are formulaic and code-like (*IMP CAES DIVI F AVGVSTVS COS XII. . .*), yet unambiguously linguistic. What prevents the Indus system from being identical?

Roman inscriptions differ from the Indus corpus in measurable ways. Individual letters repeat freely within inscriptions, reflecting alphabetic combinatorics. Abbreviations derive transparently from known words. The corpus contains inscriptions of widely varying length. Syntactic connectives reveal grammatical structure. The intra-text non-repetition finding is particularly significant: Roman inscriptions show abundant letter repetition because alphabetic writing requires it. The Indus system does not. If it includes linguistic elements, those elements must operate under combinatorial rules unlike any known writing system. This constrains H₃ without excluding it.

7 The Linear B Comparison

Linear B provides an instructive contrast. Its inscriptions include pictographic representations of recorded commodities, providing independent evidence that the texts describe inventory transactions. This correlation was crucial to Ventris's decipherment. The Indus corpus lacks this transparent correlation. No credible claim has established a link between animal motifs and sign sequences. This is why Test 2 is critical: a statistically robust animal-prefix correlation would provide the kind of independent evidence that has been missing.

The absence of longer texts is often cited against the linguistic hypothesis. But the Harappans may have written on perishable materials. What survives is the subset carved into durable materials—precisely the category most likely to serve identity-marking or administrative functions. The survival bias of the archaeological record is itself evidentially relevant.

8 The Structure of the Argument

The argument can be stated concisely. Let Q be: “Which functional architecture— H_1 , H_2 , or H_3 —does the Indus sign system instantiate?” Let S be the methodological system of statistical analysis of sign sequences. Let S' be the extended system incorporating the five tests of Section 5.

Claim 1 (Definitional Resolution): Q is well-defined. It avoids the definitional indeterminacy of the binary question by specifying three structurally distinct hypotheses with unique prediction profiles.

Claim 2 (Undecidability in S): Q is undecidable in S . The existing statistical evidence is consistent with multiple hypotheses. The Rao–Sproat exchange demonstrated this symmetry.

Claim 3 (Signal, Not Endpoint): The undecidability of Q in S signals S 's explanatory inadequacy, not Q 's inherent unanswerability.

Claim 4 (Adequate Extension): S' extends S by principled methods (archaeologically grounded predictions) without introducing the desired answer as a bare assumption.

Claim 5 (Decidability in S'): If the five tests yield clear results, the hypothesis space narrows to at most one candidate.

9 Generalizability

The dual framework—resolving definitional indeterminacy, then extending the methodological system—applies to any contested sign system: Proto-Elamite, Vinča symbols, Rongorongo. In each case, progress requires specifying what precisely is being asked, then specifying what evidence would answer it.

The broader lesson is that “Is this writing?” is not a question with a fixed answer awaiting discovery. It is a question whose *answerability* depends on both the definitional precision with which it is posed and the methodological system in which it is evaluated. Constructing the right definitions and the right system is not a preliminary to the investigation. It *is* the investigation.

10 Conclusion

The century-long debate over the Indus sign system has been impeded by two compounding failures. The first is definitional: the question was never posed with sufficient precision. The second is methodological: the statistical tools deployed lack the resources to answer it. This paper has addressed both: three precisely defined hypotheses replace the undefined binary, and five testable predictions extend the methodological system from statistics to a unified statistical-archaeological framework.

Two tests can be performed immediately with existing corpora. Three require archaeological collaboration. Together, they form a decision tree in which no two hypotheses generate identical predictions.

The existing evidence is consistently explained by H_2 , the administrative-code hypothesis. But consistency is not proof. The path forward requires specifying what would constitute proof—which is what this paper has done.

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